Depth Charge — Viva Guide (Alyan)

Common overview plus individual module questions

Course: EL-2003 COAL Lab, Fall 2026

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Read this page first. Then read your own member guide and skim the other two. The viva can ask about the whole program, not only your folder.

1. Explain the project in 30 seconds

> *Depth Charge is a single-player 80 × 25 console game in 32-bit x86 assembly. I move a submarine, fire torpedoes, and avoid falling depth charges. Some charges are hidden mines. Sonar reveals them for two seconds. The game has score, three lives, nine levels, pause, sound, and a saved high score. We used MASM, Irvine32, fixed arrays, procedures, and a simple game loop.*

Do not say it uses DOS video memory or BIOS interrupts. It is a Windows console program. Irvine32 and Windows API calls provide console input/output, timing, files, and sound.

2. The five jobs in every frame

Read keys -> Move objects -> Check hits -> Draw changed cells -> Wait

1. **Input:** `HandleInput` checks waiting keys without blocking.
2. **Update:** torpedoes move up; charges sometimes move down; a new charge may spawn; explosions age.
3. **Collisions:** compare object coordinates in memory, not screen pixels.
4. **Render:** build a complete frame in `ScreenBuf`, then compare it with `PrevBuf`.
5. **Delay:** sleep for the remaining part of the 80 ms frame target.

GameLoop in src/game.asm controls this order. gameState decides whether object updates run.

State	Value	Meaning
`ST_MENU`	0	Menu is outside the gameplay loop
`ST_PLAYING`	1	Normal updates and drawing
`ST_SONAR`	2	Objects frozen; hidden mines visible
`ST_PAUSED`	3	Objects frozen; pause message visible
`ST_GAMEOVER`	4	Save record and show final score
`ST_EXIT`	5	Leave the program

P remembers whether the game was playing or in sonar. If sonar was active, its start time is shifted forward by the paused duration, so pausing does not consume the two-second reveal. The ten-second recharge is wall-clock time and continues during an ordinary pause.

3. The one structure everyone uses

```
GameObject STRUCT
   xPos    BYTE 0
   yPos    BYTE 0
   active  BYTE 0
   hidden  BYTE 0
GameObject ENDS
```

- Size = **4 bytes**. `TYPE GameObject` is 4.
- `active = 0` means the slot is free.
- `hidden = 1` means a mine is invisible during normal drawing.
- Hidden does **not** mean inactive. Collision code still checks it.
- Torpedoes: 8 × 4 = 32 bytes.
- Charges: 15 × 4 = 60 bytes.
- Explosions: 5 × 4 = 20 bytes, plus five separate timer bytes.

Example: if ESI points to Charges[0], add esi, TYPE GameObject points to Charges[1]. Adding 1 would point inside the first object, not to the next object.

4. Screen memory is not the Windows video card

Each screen cell is one WORD:

```
low byte  = ASCII character
high byte = colour attribute
cell index = row * 80 + column
byte offset = cell index * 2
```

Example: row 5, column 10 → (5 × 80 + 10) × 2 = 820 bytes from the start of ScreenBuf.

The buffer has 80 × 25 = 2000 WORDs = 4000 bytes. This resembles old text-mode layout, but it is ordinary program memory. Gotoxy, SetTextColor, and WriteChar update the Windows console. We never write directly to B800h.

The bottom-right cell is not printed because writing there can scroll some consoles. The renderer still copies the blank cell into PrevBuf; it simply never sends that cell to the console.

5. Course-outline mapping — what we actually used

Course topic	Where to show it	Simple explanation
IDE, assemble, debug, run	`Project.sln`, Win32 build	MASM makes an OBJ; linker makes an EXE
Data definition and strings	`data.inc`, `screens.inc`	BYTE/WORD/DWORD and zero-terminated text
MOV, MOVZX, LEA	`render.inc`, `hud.inc`	Copy data, extend a byte safely, take an address

Flags and conditional jumps	input, collisions, timers	`CMP` sets flags; `JE/JB/JAE/JNZ` choose a path
Indirect/index addressing	object loops, `ScreenBuf[ebx*2]`	A register holds an address or index
Arrays and loops	torpedoes, charges, explosions	Fixed slots avoid dynamic memory allocation
Procedures and stack	every module; `USES`, `CALL`, `RET`	Reusable code; save/restore registers
Stack frame and LOCAL	`PutNumber`	Temporary digit string belongs to this call
Shift and rotate topic	`SHL` for colour, `SHR` in random	Shift bits left/right; rotate is not needed
Multiply/divide	row offset, LCG, digits, level	`MUL/IMUL` and `DIV` with correct EDX setup
String instructions	`REP STOSB/STOSW`, `REP MOVSD`	Clear arrays/buffers and copy a frame
Structures and scaling	`GameObject`, explosion indexing	`TYPE` gives the element size
Macros	`mPlaySound`	Expand a small save/Beep/restore pattern
OS services	Irvine32 and Windows API	Console, timing, files, and Beep

Not every listed course topic must appear in one project. This game does not use recursion, inline assembly, extended multiword arithmetic, or the optional REPE CMPSW renderer. Saying that honestly is better than inventing a feature.

6. Questions every member should answer

Why x86 and not x64?

The course library is Irvine32. Its procedures and library are 32-bit, so the project uses Win32 and 32-bit registers.

Why is there one .asm file but many .inc files?

INCLUDE inserts source text before assembly. It split work by owner without creating separate linking contracts. There is still one assembled source unit.

Why not call Clrscr every frame?

Clearing and redrawing everything causes flicker and unnecessary console calls. We clear only the memory buffer and print changed cells.

Why can a hidden mine hit me?

The renderer checks hidden; the collision procedure checks active and coordinates. The object remains in the array.

Why does DIV need EDX = 0 here?

Unsigned 32-bit DIV divides the 64-bit value EDX:EAX. For these small unsigned dividends, the upper half must be zero. Otherwise the result can be wrong or the quotient can overflow.

What does USES eax ecx do?

MASM generates pushes at procedure entry and matching pops before return. The caller's values in those registers are preserved. It does not automatically preserve registers not listed.

Does USES preserve flags?

No. Do not assume ZF or CF survives a procedure call unless its contract explicitly says so.

Why is the frame rate a target, not a guarantee?

The OS schedules other programs, console calls take time, and Beep blocks. The loop waits for the remaining time when possible; it cannot guarantee hard real-time deadlines.

Where is the high score saved?

In highscore.dat in the current working directory. It stores one four-byte DWORD in little-endian binary form, not readable decimal text.

What happens when I press ESC during a round?

The round ends and the menu returns. The current score is not saved. Saving happens at game over, as described in the instructions.

7. Three-minute demonstration script

6. Show the menu and high score. Open instructions, return, then start.
7. Move with both letters and arrow keys. Touch both boundaries.
8. Fire several torpedoes. Point out the small firing gap.
9. Let charges fall. Explain that `*` is visible and hidden mines are in memory.
10. Press `SPACE`. Point to `SONAR ACTIVE` and any green `!` marks.
11. Press `P` during sonar, wait, then resume. Explain the remembered state.
12. Hit a charge if possible; show the explosion and score change.
13. Lose three lives or use the debugger to demonstrate `LoseLife` honestly.
14. Show game over and the saved high score. Restart the program to show persistence.
15. Start a new round and show score 0, lives 3, level 1, and empty arrays.

If random play does not produce a hit during the short demo, use a prepared debugger state and say it is a controlled test. Do not pretend a random event was guaranteed.

8. Debugger questions you may be asked

- Put a breakpoint at `DoPlaying` and step through one frame.
- Watch `subX`, `score`, `lives`, `level`, and `gameState`.
- Open Memory at `Charges` and identify four bytes per object.
- Open Memory at `ScreenBuf` and find a known character/colour WORD.
- Break at `Hit` in `collision.inc` and explain which two objects matched.

- Break at `Rand` and show `EDX` cleared before `DIV`.
- Break at `SaveHighScore` and show the four-byte write.

9. Honest preparation checklist

- [ ] I built the exact commit on my own/lab Windows PC.
- [ ] I can draw the frame loop without looking.
- [ ] I can explain every procedure in my own folder.
- [ ] I read the other two members' summaries.
- [ ] I ran the manual tests and recorded real results.
- [ ] I personalized my report and disclosed AI assistance.
- [ ] I can explain a limitation instead of guessing.

Viva guide — Syed Alyan Hussain (25K-0648)

Your area: console setup, screen buffers, changed-cell rendering, HUD, explosions, sonar.

Files: src/alyan/.

Read 00-common.md first. Be ready to explain both your drawing code and why hidden objects still collide.

1. Your 30-second introduction

> *I draw the game without clearing the whole console every frame. BuildFrame paints objects into a WORD array. Render compares it with the previous frame and prints only changed cells. I also draw the status bar, animate explosions, and control the sonar reveal and recharge.*

2. Procedure map

Procedure	Input	Result
`InitSystem`	None	Seeds random, sets title, hides cursor
`PutCell`	`DH/DL` row/col, `AL` char, `AH` colour	Writes one WORD into `ScreenBuf`
`PutString`	`ESI` string, `DH/DL`, `AH` colour	Writes text into the buffer
`ClearBuffer`	None	Fills current buffer with blank WORDs
`InvalidateScreen`	None	Fills previous buffer with `FFFFh`
`DrawBorders`	None	Adds top/bottom ASCII borders
`BuildFrame`	Shared game data	Complete next frame in memory
`Render`	Two buffers	Prints changed cells; copies current to previous
`PutNumber`	`EAX` value, `DH/DL`, `BL` colour	Five decimal characters in the buffer
`DrawHUD`	Score/lives/level/state	Status row and pause message

`StartExplosion`	`AL` col, `AH` row	Finds a free effect slot, timer = 5
`UpdateExplosions`	Effect timers	Decrements timers and frees expired slots
`DrawExplosions`	Effect slots	Chooses character/colour from lookup tables
`StartSonar`	Ready request from input	Sets sonar state and start time
`UpdateSonar`	Start time	Plays sweep once; ends after about 2 s; sets recharge

3. Buffer example to memorize

ScreenBuf and PrevBuf each contain 2000 WORDs.

row 5, column 10
index = 5*80 + 10 = 410
byte offset = 410*2 = 820

If AL='U' and AH=ATTR_SUB, PutCell stores both in one WORD. MOVZX converts the row/column bytes to 32-bit values before arithmetic. IMUL ebx,ebx,80 calculates the row start. Scale *2 changes a cell index into a byte offset.

PutCell also checks row < 25 and column < 80, so a bad coordinate cannot write outside the buffer.

4. Why two buffers reduce flicker

16. `ClearBuffer` clears memory, not the console.
17. `BuildFrame` draws HUD, borders, submarine, torpedoes, visible charges, and explosions.
18. `Render` compares each cell with `PrevBuf`.
19. Equal cells are skipped.
20. Changed cells use `Gotoxy`, `SetTextColor`, and `WriteChar`.
21. `REP MOVSD` copies the complete current buffer to the previous buffer.

There are 2000 WORDs but only 1000 DWORD copies because one DWORD contains two WORDs.

Why FFFFh in InvalidateScreen? After a menu or Clrscr, the old buffer no longer describes the console. FFFFh forces every normal cell to look changed on the next render.

Why skip the last cell? Printing row 24, column 79 can scroll the console. The border and control text stop before it.

5. HUD and decimal conversion

The HUD shows score, lives, level, and sonar status. PutNumber converts a number to five digits:

22. Save `EDX` because it holds the column and `DIV` will change it.
23. Divide by 10.
24. Remainder is the last digit; add `'0'` to make ASCII.

25. Store digits from right to left in a six-byte `LOCAL` buffer.
26. Restore the position and call `PutString`.

Example: 123 → digits '0' '0' '1' '2' '3' followed by zero terminator.

LOCAL digits[6]:BYTE creates temporary stack space. MASM uses a stack frame (EBP) for the procedure. This is a small, real use of the course's stack-frame topic.

The HUD is designed for scores up to 99999. The score variable itself is a DWORD; the simple five-digit display is a stated presentation limit.

6. Explosion trace

StartExplosion receives column in AL, row in AH. It searches five slots. If all are busy, only the visual effect is skipped; the hit and score still happen.

Timer values and characters:

Timer	Table index	Character
5	4	`@`
4	3	`#`
3	2	`*`
2	1	`+`
1	0	`.`

The game updates explosions before collision checks. A new hit draws timer 5 in that frame, then four more frames count down to 1. The next update frees the slot. Explosions freeze with the rest of the objects during sonar/pause.

[esi+ebx*TYPE GameObject] demonstrates base-plus-index-scaled addressing. TYPE GameObject is 4.

7. Sonar trace

27. Input accepts `SPACE` only in normal play and only when ready.
28. `StartSonar` sets `ST_SONAR`, saves the start time, clears `sonarPlayed`.
29. The loop skips object updates.
30. The first `UpdateSonar` builds and renders the revealed frame, then plays the rising sweep once.
31. Hidden charges are drawn as green `!` only while the state is sonar.
32. After about 2000 ms, state returns to playing and `sonarReadyAt = now + 10000`.
33. The HUD shows `ACTIVE`, whole seconds remaining, or `READY`.

The sweep is about 250 ms and is included in the two-second interval because the start time is saved before the sound. Timing is checked at frame boundaries, so it is approximate, not a hardware guarantee.

Pause fix: Anas's input code shifts sonarStart forward by the paused duration. Your UpdateSonar then naturally gives the remaining reveal time.

8. Likely questions and short answers

Is this double buffering in the graphics-card sense? It is the same idea—build then present—but implemented with console character cells, not a GPU swap chain.

Why not optional RenderFast? The ordinary comparison loop is easier to debug and explain. The proposal's core requirement is changed-cell drawing, which this provides.

Where is the hidden flag checked? In BuildFrame. Collision code does not use it to decide whether an object exists.

Why does REP MOVSD use 1000, not 2000? It copies DWORDs. Two WORD cells fit in one DWORD.

What happens if ClearBuffer forgets CLD? If the direction flag were set, EDI would move backward and corrupt memory. CLD guarantees forward movement.

Why does GetMseconds require resetting AH before text? A procedure may change EAX; do not assume a colour previously placed in AH survives.

Why can't you draw during ST_PAUSED and skip input? The player must still be able to resume or exit. The loop skips updates, not input.

9. Practical task before viva

- Show the WORD at row 5, column 10 in Memory.
- Break at `Render` and show an unchanged cell being skipped.
- Change `score` to 123 in the debugger and explain the five HUD characters.
- Start an explosion and watch its timer go 5 → 0.
- Activate sonar with a hidden charge present and point to the `hidden` branch.
- Explain why the bottom-right cell remains blank.

10. Limitations you can state confidently

- The console must be at least 80 × 25; the program draws that area rather than implementing resizable graphics.
- Five HUD digits are a simple display choice.
- Timing depends on the OS frame loop.
- The optimized `REPE CMPSW` version is intentionally absent.